

Midterm Exam -1

CS 101 Algorithmic Problem Solving

Fall 2023

Name(s): _____

HU ID (e.g., xy01042): _____

1. Fast car

You have decided to place Pakistan on the map of fast car manufacturers. Your team is going to build a car that weighs lb pounds and goes from 0 to 60 mph in t seconds. Write a function, `engine_force()`, that takes lb and t as parameters and prints the force in Newtons that the engine must produce.

Constraints

- 1 mile = 1.6 km = 1600 m.
- 1 pound = 0.45 kg.
- Acceleration: $v = u + a \times t$
- Force: $F = m \times a$

Interaction

The input comprises of two space separated integers denoting the value of lb, t .

The output must contain an integer denoting the computed value of force.

Sample

Input	Output
1500 5	3621
2000 6	4023

In the first case, the car weighs 1500 pounds and accelerates from 0 to 60 mph in 5 seconds, and the engine must produce 3621 N of force.

In the second case, the car weighs 2000 pounds and accelerates from 0 to 60 mph in 6 seconds, and the engine must produce 4023 N of force.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1800 4	
1000 7	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any two of the inputs provided in the Exercise section above, dry run your psuedocode in the space below.

2. Chess Master

Impressed with the inventor of chess, an Indian rajah asks the inventor how they wish to be rewarded. The sly inventor asks for grains of rice to be placed on the chessboard in such a way that one grain is placed on the first square, two grains on the second square,

four grains on the third square, eight grains on the fourth square, and so on, doubling the previous number each time, until all 64 squares are filled. The rajah, thinking the reward to be trivial, agrees to the inventor's request. Little did the rajah realize that his granaries would be empty well before he filled the final square.

Write a function `grains_placed()` that takes the number of squares, `squares`, as a parameter and returns `grains`, the total number of grains of rice to be placed on the first n squares of the board. Make sure to call your function in the pseudocode section.

Constraints

- $n > 1$

Interaction

The input comprises a single line containing n .

The output must contain an integer denoting the total number of grains of rice placed on the chessboard using the given doubling pattern.

Sample

Input	Output
5	31
8	255

In the first case, 31 total grains are placed on 5 squares of the board.

In the second case, 255 total grains are placed on 8 squares of the board.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
6	
4	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

3. Pyramid of stars

Write a function called *pyramid(n)* that generates and prints a pattern of stars forming a pyramid. Each row of the pyramid should be indented appropriately.

Make sure to call your function in the pseudocode section.

Constraints

- $1 \leq n \leq 100$

Interaction

The input comprises a single line containing an integer n .

The output must contain a pyramid of stars indented to the right S .

Sample

Input	Output
4	<pre> * ** *** ****</pre>
3	<pre> * ** ***</pre>

In the first case, a right aligned pyramid is created with 4 rows.

In the first case, a right aligned pyramid is created with 3 rows.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
1	
6	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any two of the inputs provided in the Exercise section above, dry run your pseudocode

in the space below.

4. Tomorrow!

In a quiet village nestled between rolling hills, a peculiar event unfolded. Every evening, the village folk gathered to witness a gifted scholar, adept in the ways of time and dates, predict the immediate successor of each day. As curiosity and wonder fill the air, your task is to emulate this mystical talent by creating a function called `next_day()`. This function, akin to the village scholar's predictions, will determine the immediate successor of a given date provided in numeric form, offering a glimpse into the village's timeless traditions.

Write a function `next_day()` that takes a date and returns its immediate successor. The date will be entered in numeric form with three separate input parameters: `year`, `month`, and `day`.

Constraints:

- $year, month, day \in \mathbb{Z}$
- $1 \leq year \leq 9999$
- $1 \leq month \leq 12$
- $1 \leq day \leq 31$

Interaction

Input comprises of three space separated integers, `year` (a positive integer representing the year), `month` (an integer from 1 to 12 representing the month), and `day` (an integer representing the day of the month).

Output must be a string representing the immediate successor date in the format "YYYY-MM-DD".

Sample

Input	Output
2023 10 18	2023-10-19
2013 12 31	2014-01-01

In the first case, the input date is October 18th, 2023. The output date is for the next day, which is October 19th, 2023.

In the second case, the input date is November 30th, 2022. The output date is it's immediate successor, December 1st, 2022.

Exercise

In the space provided, indicate the outputs for the given inputs.

Input	Output
2022 11 30	
2024 02 28	

Problem Identification

Briefly explain the underlying problem you identified in the above question that led you to your solution.

Pseudocode**Dry Run**

Using any one of the inputs provided in the Exercise section above, dry run your pseudocode in the space below.

5. Temperature Confusion

Fatima lives in Pakistan but her best friend Sana lives in United States. They both use different temperature system as United states doesn't use Celsius. Therefore, whenever they talk about weather there is great confusion. Fatima decides to make a function that takes temperature T and a letter S that indicates the unit temperature needs to be converted to.

Help her develop such a function called TempConverter which uses two helper functions to print the converted temperature and the system it is converted to. If $S = C$ then FahrenheitToCelsius is used and if $S = F$ then CelsiusToFahrenheit is used. Helper functions just return the converted temperature. Make sure to call the main function to print the output.

Hint: Temperature in degrees Fahrenheit = (Temperature in degrees Celsius * 9/5) + 32.

Constraints

- $S = C$ or $S = F$
- $0 \leq T \leq 212$
- Round off the temperature to 2 decimal places.

Interaction

The input comprises a single line containing an integer denoting the value of T and a letter denoting the value of S .

The output must contain a line stating the converted temperature and the system it is converted to.

Sample

Input	Output
44 C	6.67 Degree Celsius is the temperature for Fatima.
0 F	32.00 Fahrenheit is the temperature for Sana.

In the first case, 44 Fahrenheit is to be converted to Celsius. Using the formula and rounding it off we get 6.67. Moreover, the conversion is to the degree Celsius hence, it is for Fatima.

In the second case, 0 Degrees Celsius is to be converted to Fahrenheit. The given formula is used to obtain the temperature in Fahrenheit and as it is not in Celsius we know it is for Sana.

Proposed Solution

```
def FahrenheitToCelsius(T):  
    return round((T - 32) * 5/9, 2)  
  
def CelsiusToFahrenheit(T):  
    return round((T * 9/5) + 32, 2)  
  
def TempConverter(T, S):  
  
    if S == 'C':  
        converted_temp = CelsiusToFahrenheit(T)  
        system = "Degree Celsius"  
        print(str(converted_temp) + " " + system + " is the temperature for Fatima.")
```

```
elif S == 'F':  
    converted_temp = FahrenheitToCelsius(T)  
    system = "Fahrenheit"  
    print(str(converted_temp) + " " + system + " is the temperature for Sana.")
```

Dry Run

Using the inputs provided in the Sample section above, dry run the proposed pseudocode in the space below to find out errors.

Error Identification

Briefly explain the errors you identified in the proposed code solution. Mention the line number and errors in each line.

Correct Solution

Rewrite the lines of code you mentioned above with their errors corrected.

Rough Work